

KUBeetle-S: An insect-like, tailless, hover-capable robot that can fly with a low-torque control mechanism

International Journal of Micro Air Vehicles

Volume 11: 1–10

© The Author(s) 2019

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/1756829319861371

journals.sagepub.com/home/mav



Hoang Vu Phan^{1,2} , Steven Aurecianus³, Taesam Kang³ and Hoon Cheol Park^{1,2}

Abstract

For an insect-like tailless flying robot, flapping wings should be able to produce control force as well as flight force to keep the robot staying airborne. This capability requires an active control mechanism, which should be integrated with lightweight microcontrol actuators that can produce sufficient control torques to stabilize the robot due to its inherent instability. In this work, we propose a control mechanism integrated in a hover-capable, two-winged, flapping-wing, 16.4 g flying robot (KUBeetle-S) that can simultaneously change the wing stroke-plane and wing twist. Tilting the stroke plane causes changes in the direction of average thrust and the wing twist distribution to produce control torques for pitch and roll. For yaw (heading change), root spars of left and right wings are adjusted asymmetrically to change the wing twist during flapping motion, resulting in yaw torque generation. Changes in wing kinematics were validated by measuring wing kinematics using three synchronized high-speed cameras. We then performed a series of experiments using a six-axis force/torque load cell to evaluate the effectiveness of the control mechanism via torque generation. We prototyped the robot by integrating the control mechanism with sub-micro servos as control actuators and flight control board. Free flight tests were finally conducted to verify the possibility of attitude control.

Keywords

Flapping wing micro aerial vehicle, stroke plane modulation, control mechanism, biomimetics, insect flight

Received 12 February 2019; accepted 7 April 2019

Introduction

In the absence of tail control surface, insects can modify their wing kinematics to produce control force for attitude change during flight.^{1,2} In particular, shifting the stroke plane position or flapping angle range for pitch response has been found in fruit fly, *Drosophila melanogaster*,³ and hoverflies, *Eristalis tenax* and *Episyrphus balteatus*.⁴ For roll and yaw responses, many insect species such as *Drosophila* spp.,⁵ *Musca domestica*,⁶ *Calliphora* spp.,⁷ dragonflies,⁸ beetles,⁹ and hawkmoths¹⁰ can change their flapping amplitudes of two wings.

Mimicking these complex kinematics and manipulation abilities is a challenging task for developing lightweight insect-like tailless flapping-wing flying robots. Without tail stabilizer, the main flapping wings should be incorporated with a control mechanism to produce control force as well as sufficient flight force to

keep the robot staying airborne. Due to a limited take-off weight of the robot, a proposed control mechanism should be fitted for lightweight actuators, which generate a low actuation torque. Additionally, the mechanism needs to produce enough control torques for attitude changes. Due to these hurdles, only a few

¹Artificial Muscle Research Center, Konkuk University, Seoul, South Korea

²Department of Smart Vehicle Engineering, Konkuk University, Seoul, South Korea

³Department of Aerospace Information Engineering, Konkuk University, Seoul, South Korea

Corresponding author:

Hoon Cheol Park, Artificial Muscle Research Center, Konkuk University, 120 Neungdong-ro, Gwangjin-gu, Sanghuh Research Institute Building, Room 502, Seoul 05029, South Korea.

Email: hcpark@konkuk.ac.kr



hovering tailless flying robots are ready for free flight, although many research groups have successfully developed bird-like tailed robots.^{11–15} A 19 g Nano Hummingbird developed by AeroVironment¹⁶ is the first tailless two-winged robot that can successfully perform stable controlled flight. It utilizes a wing twist modulation mechanism by controlling wing root spars symmetrically or asymmetrically for roll, pitch, and yaw controls. Similar wing twist modulation mechanism can be found in a 21 g KUBeetle¹⁷ and a 22 g Colibri robot.¹⁸ Otherwise, wing kinematics modulation has been used as a control approach in a tiny 80 mg Robobee¹⁹ and a 12 g robotic hummingbird.²⁰ In this approach, wing stroke amplitudes of the left and right wings are changed asymmetrically for roll control. By shifting the mean stroke angle of the two wings forward or aft, pitch control is obtained. In addition, modulating stroke velocities in each half-stroke within a flapping cycle can result in a yaw torque generation for yaw control. By mimicking flight control strategies of fruit fly, Delfly Nimble²¹ has performed agile flight by manipulating flapping frequencies of the left and right wings for roll control. Meanwhile, pitch and yaw controls were obtained by shifting the mean stroke angle and modulating wing root spars of the two wings asymmetrically, respectively. Another flight control approach is by modulating the wing stroke-plane for pitch and yaw controls. This approach has been used in a 62 g Robotic Hummingbird²² and a recently released X-wing (four-wing) flying robot, NUS-Roboticbird.²³

We have also developed a tailless flying robot KUBeetle that can perform stable flight for about 40s.¹⁷ We have tested many control approaches including modulation of mean stroke angle,²⁴ modulation of stroke-plane angle,²⁵ and modulation of wing twist.¹⁷ However, these control mechanisms were complicated for fabrication and high-torque actuators were required for implementation. In this work, we proposed a simple and lightweight control mechanism that could modify the stroke plane and wing twist synchronously for attitude changes. We designed and fabricated the mechanism and evaluated its control force and torque generation using a six-axis force/torque load cell. The control mechanism actuated by three micro servos was then integrated in the flying robot for free controlled flight test.

Flapping-wing mechanism

The tailless robot is developed aiming to mimic the flight of horned beetle (*Allomyrina dichotoma*) known to be capable of hovering flight.²⁶ Since the mechanical design of the flapping-wing mechanism was presented in detail in ref.,¹⁷ this paper shows only a brief summary.

Two deformable wings are actuated by a 3.5 g coreless motor (Chaoli CL720, China) through a gearbox (24:1) to amplify the torque of the motor and a transmission linkage system for converting rotary motion to flapping motion using a combined operation of a four-bar linkage and a pulley-string mechanism. Wing with a length of 75 mm was made of 10 μ m Mylar film as wing membrane and one-layer carbon strips as reinforced veins. After installing to the flapping mechanism, these two wings can deform, creating spanwise twist and chordwise camber during the flapping motion.¹⁶ Flapping amplitude is approximately 190° allowing the presence of clap-and-fling effect for vertical force augmentation.²⁷ Effect of vein structures on force generation and power requirement was also investigated previously.²⁸

Attitude control mechanism

Conceptual design and fabrication

Figure 1 shows CAD images of control torque generation mechanisms for pitch, roll, and yaw attitude changes. The flapping-wing mechanism (including the motor) is able to rotate about two hinges (H1 and H2 in Figure 1) for pitch and roll controls. The rotating hinges are designed to locate at about the center of gravity (CG) of the flapping-wing mechanism. Thus, during rotation, the CG of the robot is not significantly changed. By tilting the flapping-wing mechanism around the hinge H1 forward or backward (Figure 1, side view), the wing stroke plane is tilted in the same direction to the change in the direction of force for pitch torque generation, as shown in Figure 2(a). Additionally, since the wing root spars are constrained, in this case, tilting the stroke plane causes modulation in wing twist resulting in change of magnitude of force to produce additional pitch torque (Figure 2(a)). To generate roll control torque, the flapping-wing mechanism is rotated about hinge H2 to tilt the stroke plane laterally to the left or right (Figure 1, back view). As a result, wing twists of the left and right wings are also modulated asymmetrically to change the magnitude of forces in the two wings, as depicted in Figure 2(b). Thus, pitch and roll control torques are generated by simultaneously changing the stroke plane and modulating the wing twist. Yaw motion is controlled by rotating the root spars of the left and right wings in the opposite directions, as shown in Figure 1 (Isometric view). Modulation in the root spars changes the angle of attack (AoA) during flapping motion resulting in production of asymmetric horizontal forces in the two wings to generate yaw torque (Figure 2(c)).

Based on the conceptual design, we fabricated parts for the control mechanism using a 0.8 mm

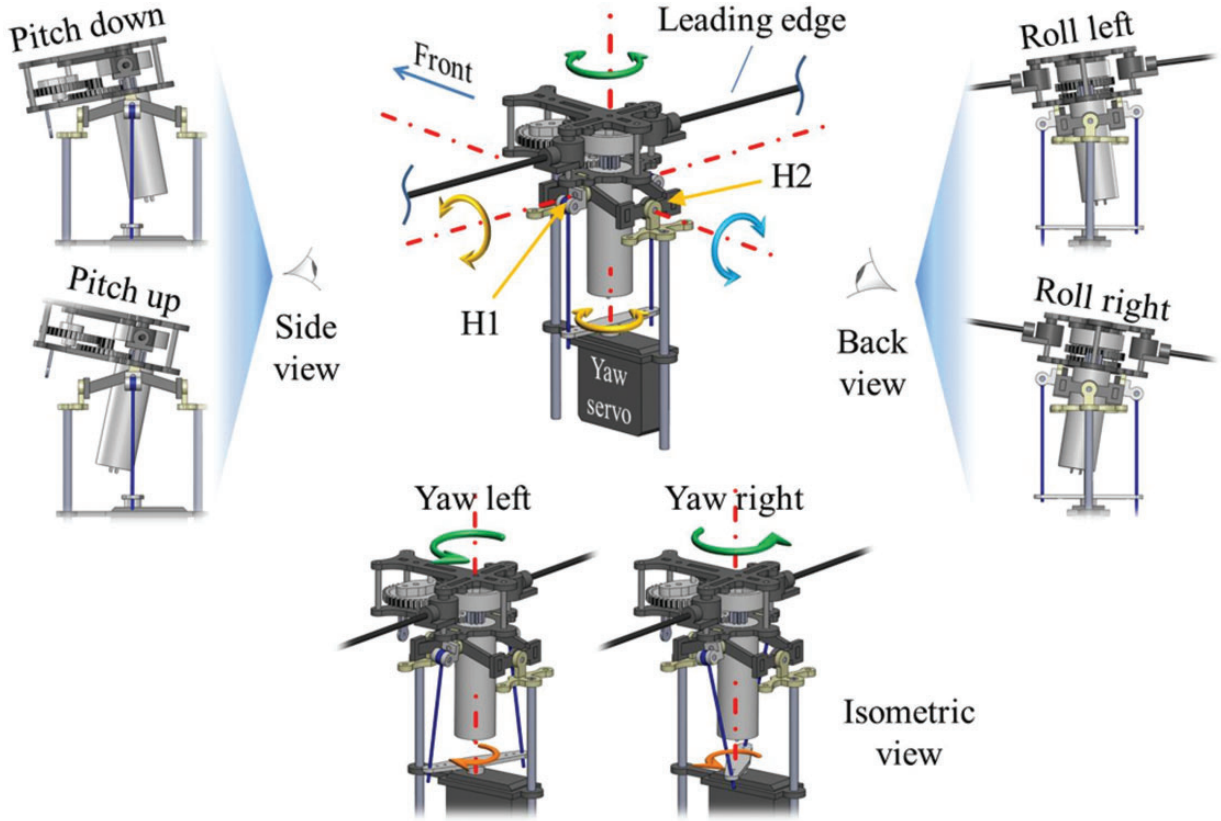


Figure 1. Conceptual design of the control mechanism for attitude changes.

carbon/epoxy panel and installed it in the flying robot as depicted in Figure 3. Pitch and roll controls are actuated independently by two sub-micro LZ servos (0.5 g, microflerradio.com). We tested and found that the sub-micro servos could produce enough torque to tilt the stroke plane for attitude control and stability during flapping motion. Meanwhile, yaw is controlled by modulating wing root spars, which are directly affected by high oscillating flapping forces. Therefore, yaw control mechanism is actuated by a conventional high-torque digital servo (HK-5320, hobbyking.com) weighing 1.5 g.

Wing kinematics

We used three synchronized high-speed cameras (Photron Ultima APX, 1024×1024 pixels, 2000 fps) to capture the wing motion, as shown in Figure 4. The synchronized 2D images captured by the three cameras were inputted to the DLTdv3 program²⁹ to obtain 3D coordinates of markers placed on the wing, which are referenced from the coordinates of a calibration frame (Figure 4). These 3D coordinates of markers were then converted into sweep angle ψ (the angle between the x -axis and the wing leading edge)

and wing rotational angle θ_r (the angle between the wing chord and the stroke plane) (Figure 4). Procedure to obtain the wing kinematics can be found in more details elsewhere.^{30–32}

The flapping wing was operated at a flapping frequency of about 23 Hz, which is expected to produce sufficient lift for flight. Because the left and right wings were designed to flap symmetrically, only the right wing was captured for wing kinematics analysis. Figure 5 shows sweep angles and wing rotational angles at 50% wingspan averaged from three flapping cycles for different control inputs of hovering, pitch, roll, and yaw motions. As shown in Figure 5(a), sweep amplitudes are similar for different control inputs. However, time durations of the downstroke and the upstroke are slightly different, especially for pitch and yaw controls due to variation of AoA in each stroke, as shown in Figure 2. Figure 5(b) shows wing rotational angle when the stroke plane is tilted $\pm 15^\circ$ for pitch control. Because the wing root is constrained, change of the stroke plane causes variation in the wing rotational angle or AoA during downstroke and upstroke for the pitch control inputs, compared to the hovering case. Figure 5(c) plots rotational angles for roll control. For roll-right control, the stroke plane

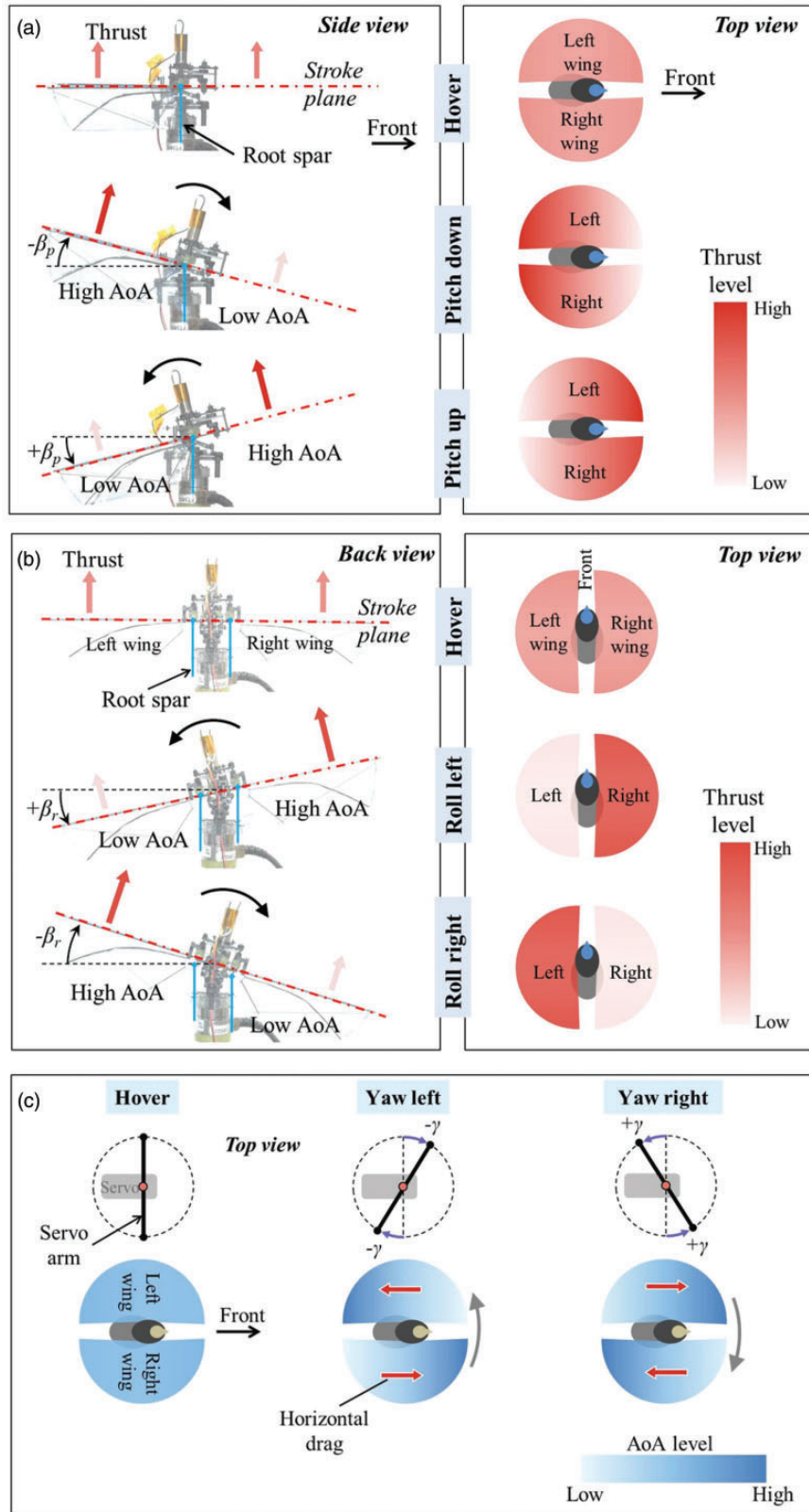


Figure 2. Working principles of the designed control mechanism for: (a) pitch, (b) roll, and (c) yaw controls.

was tilted to the right, causing decrease and increase of AoAs in right and left wings, respectively. By tilting the stroke plane to the left, roll left can be achieved. Figure 5(d) plots wing rotational angles for yaw control when the yaw servo's arm is rotated for $\pm 20^\circ$. We can see that time histories of the rotational angles for pitch (Figure 5(b)) and yaw (Figure 5(d)) controls are similar. This is because wing roots are under the same manner of deflection with respect to the stroke plane. Thus, tilting the stroke plane causes change in the wing rotational angle or AoA, supporting the production of control torque for attitude change.

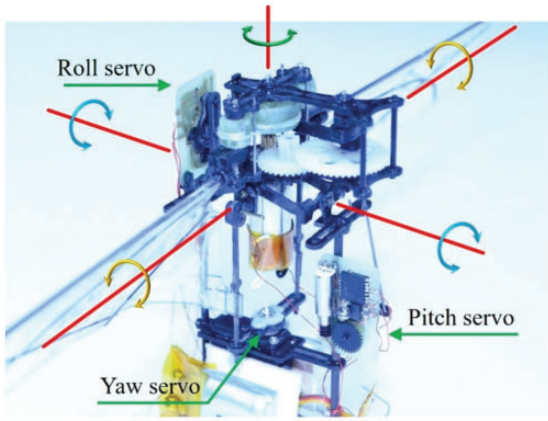


Figure 3. Prototype of the control mechanism integrated in the flying robot.

Force and torque generation

To investigate the capability of control torques generation, we set up an experiment using a six-axis load cell (Nano 17, ATI Industrial Automation, USA, force resolution ≈ 0.3 gf, torque resolution ≈ 0.0156 N·mm), as shown in Figure 6. We located the load cell close to the location of the robot's center of gravity (CG). Thus, torques obtained from the load cell can be regarded as those about the CG. We excited the flapping-wing system using an external power supply (E36103A, Keysight, Korea) at a flapping frequency of 23 Hz.

Figure 7 shows forces and torques generated in the three axes for different pitch, roll, and yaw inputs. At 23 Hz, a mean vertical force (F_z) of about 18.5 gf was produced. Changing control inputs (β_p) for pitch resulted in linear changes in the horizontal force (F_x) and pitch torque (M_y). In the range of control input (β_p) from -15° (pitch down) to 15° (pitch up), the measured F_x varied from -4.0 gf to 3.1 gf and M_y varied from -4.2 N·mm to 2.7 N·mm. Without control inputs for hovering condition ($\beta_p = 0^\circ$), pitch torque of about -0.5 N·mm was generated. This is due to slight vertical misalignment between the mean aerodynamic force center and the load cell. Effect of pitch control inputs on other forces (F_y and F_z) and torques (M_x and M_z) was insignificant. For roll control, changing the control input (β_r) from -15° (roll right) to 15° (roll left) resulted in changes of lateral force (F_y) from 2.4 gf to -2.6 gf and roll torque (M_x) from -2.8 N·mm to 2.6 N·mm.

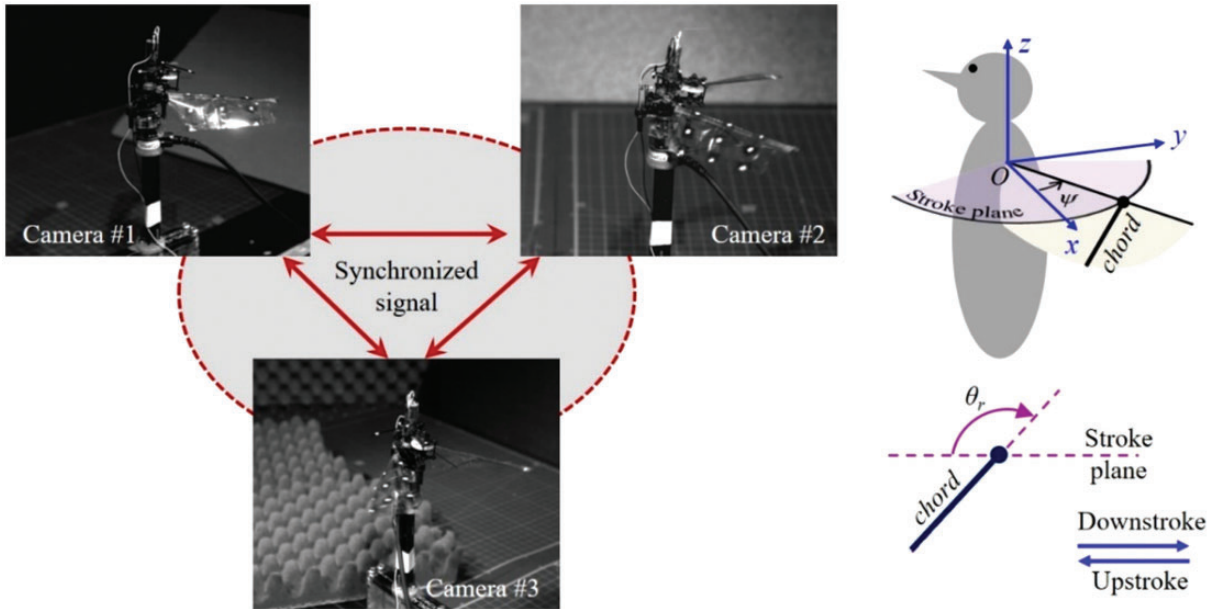


Figure 4. Measurement of wing kinematics using three synchronized high-speed cameras and definition of sweep angle (ψ) and rotational angle (θ_r).

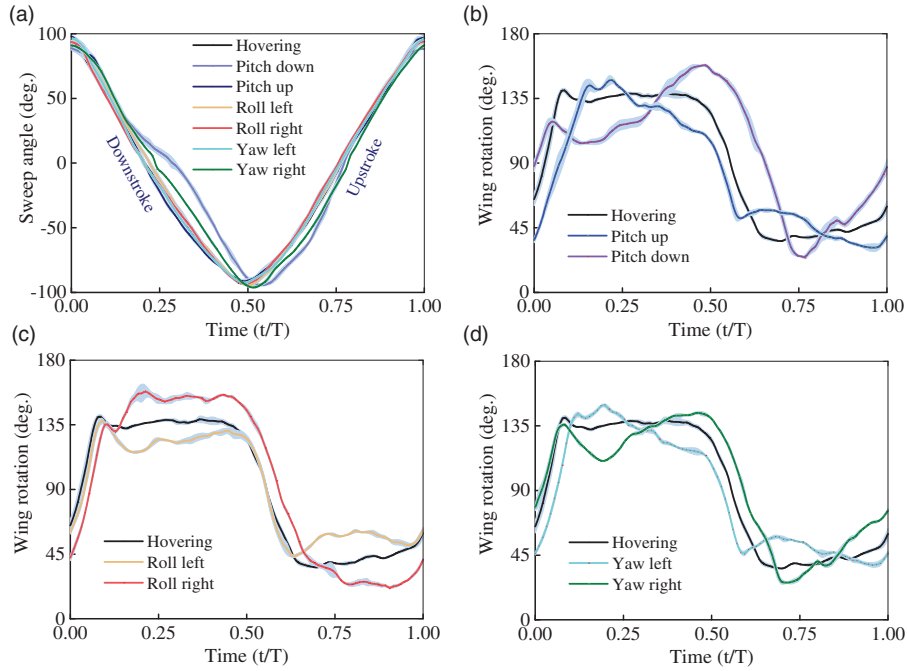


Figure 5. Time histories of sweep angle and wing rotational angle at mid-span ($0.5R$) of the right wing: (a) sweep angles, and (b–d) wing rotational angles during (b) pitch control, (c) roll control, and (d) yaw control.

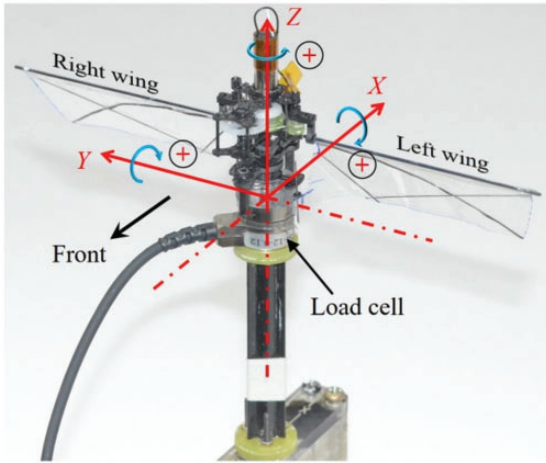


Figure 6. Experimental setup for forces and torques measurement using the load cell Nano 17.

A slight decrease in the vertical force was found for a higher roll command angle. Yaw torque (M_z) is less sensitive to the yaw command (γ) compared to those of pitch and roll commands. For the range of yaw inputs by the tilt of yaw servo's arm (Figure 2) from -20° to 20° , yaw torque varied from 0.5 N·mm to -0.3 N·mm (a reference of 0.06 N·mm at hovering condition). Thus, the proposed control mechanism can produce reasonable control torques to be implemented in the flying robot.

Attitude stabilization

For attitude control and stabilization, a custom-built 1 g, four-layer control board (23 mm length $\times$ 17 mm width) was mounted onboard. The board consists of a microprocessor ARM Cortex-M4 32-bit STM32L432KC, a 9-axis MPU-9250 consisting of three-axis gyroscope, accelerometer and magnetometer (AK8963C), a 2.4GHz transceiver nRF24L01+, and power regulators, as shown in Figure 8(a). Due to magnetic field interference produced by surrounding high current lines, servos, and DC motor, we used an extensional magnetometer located at the tip of front leg apart from interference sources (Figure 9) instead of the one integrated in the control board to ensure stable measurement.

In addition, we custom built a remote control with the same transceiver as control board and similar parts used in a conventional remote control such as joysticks, potentiometers, and buttons, as shown in Figure 8(b). Furthermore, an IMU was added to optionally allow control by motion when selected. The remote control works as control command transmitter and flight data receiver at the same time. Flight data are sent to a computer in real time for observation and recording. Data rate depends on radio frequency signal stability and data size, which can be modified accordingly. With 14-byte payload data transmission and 115,200 baud rates UART/USART communication to the computer, the data rate is around 100 Hz.

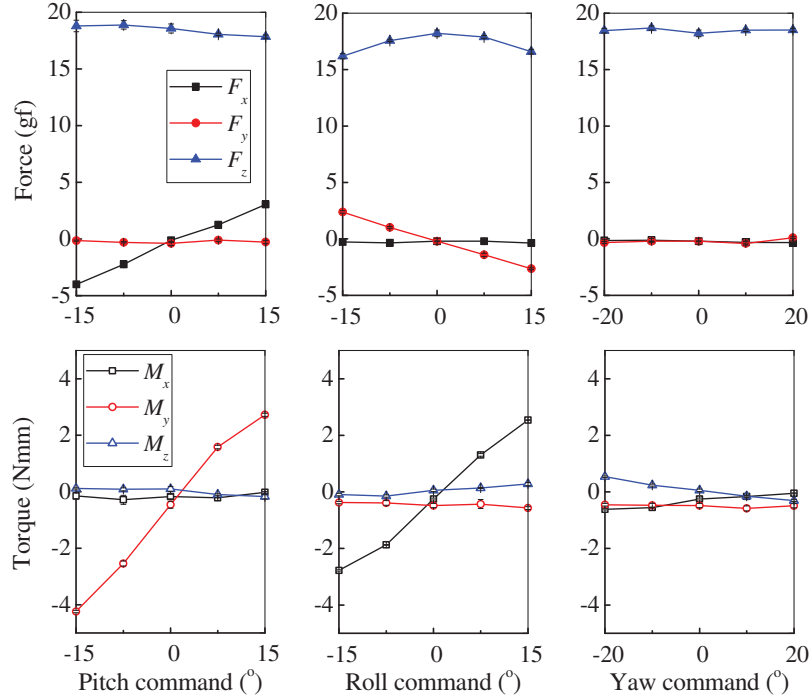


Figure 7. Generation of forces and torques in three axes for a range of control inputs.

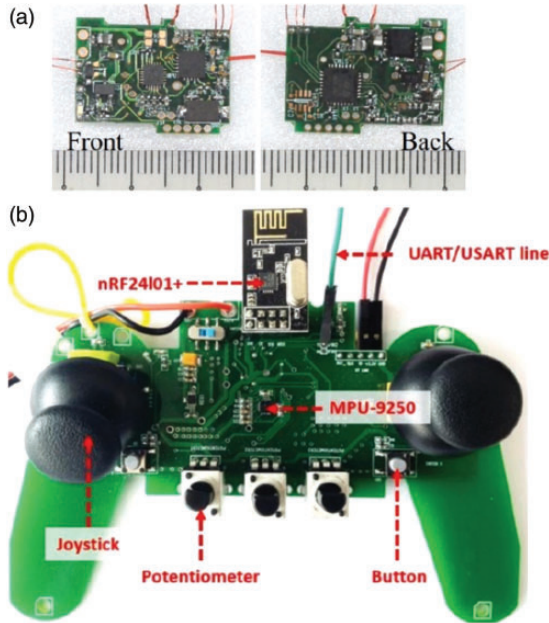


Figure 8. (a) Custom-built flight control board used in the flying robot and (b) remote control.

Due to inherent instability of tailless robot, a feedback proportional-derivative controller was implemented in the control system to sense the attitudes of the robot that could be determined by roll, pitch, yaw Euler angles (ϕ , θ , ψ , respectively) and angular rates

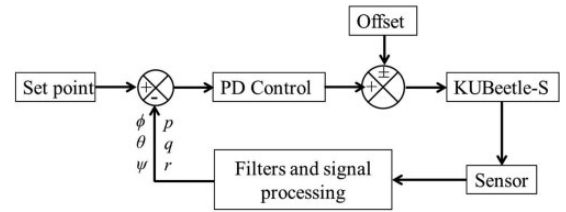


Figure 9. Attitude control loop for KUBeetle-S robot.

(p , q , r , respectively) about the X -, Y -, and Z -axes, respectively (Figure 9). Accelerometer readings are used to obtain roll and pitch angles. Meanwhile, angular rates are estimated from gyroscope readings. Thus, roll and pitch attitudes can be estimated either by accelerometers or gyroscopes. However, data obtained from accelerometers are strongly affected by vibrations caused by a high flapping motion. On the other hand, gyroscope signals are less sensitive to disturbances, but drift by time. To solve this issue, a combination of low-pass and Kalman filters that could use signals from both gyroscope and accelerometer was used to filter the roll and pitch angles, while a low-pass filter was used to smooth the yaw signal.³³

Flight experiment

The robot named KUBeetle-S with a wingspan of 170 mm and height of 75 mm (Figure 10) weighs

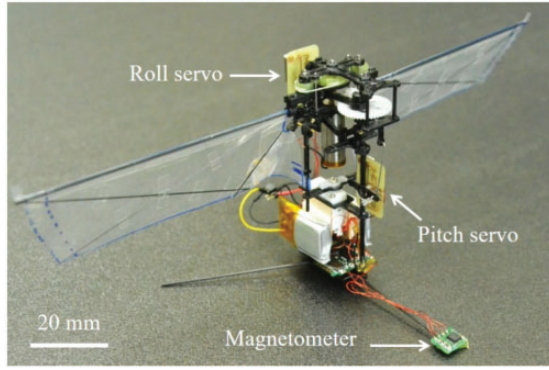


Figure 10. KUBeetle-S integrated with an onboard control system.

Table 1. Mass breakdown of the KUBeetle-S.

Component	Weight (g)	Percentage (%)
Flapping mechanism	2.6	15.9
Driving motor	3.4	20.7
Control servos	2.6	15.9
Control mechanism	0.7	4.3
Batteries	4.0	24.4
Control board	1.5	9.1
Supporting frames	0.9	5.5
Wings	0.4	2.4
Wires	0.3	1.8
Total mass	16.4	100

about 16.4 g, as shown in Table 1. The driving motor was powered by a two-cell lithium-polymer battery connected in series (7.4 V, 70 mAh). With onboard regulators in the control board, the servos and MPU9250 sensor were supplied by 5 V and 3.3 V sources, respectively. The power inputted to the motor was manually controlled through the power throttle stick.

The KUBeetle-S was demonstrated for free flight to evaluate the effectiveness of the control mechanism. Figure 11 shows composite images of the robot during its flight in the first 30 s after takeoff from the ground. With the battery capacity, the robot demonstrated its successful flight for about 3 min. Figure 12 plots body attitude without heading hold activation. In this case, pitch (θ) and roll (ϕ) angles showed small variations from the reference of 0° . However, since yaw motion was stabilized using gyroscope signal only, it experienced drift during flight. Additionally, yaw control torque was less sensitive to the yaw input, as shown in Figure 7. Therefore, heading direction (ψ) drifted during flight. However, the upright stability of the robot was unaffected by the heading stability. On the other hand, with activation of heading hold using sensing signals from the magnetometer, heading direction of the robot was maintained,



Figure 11. Composite images of KUBeetle-S during hovering flight.

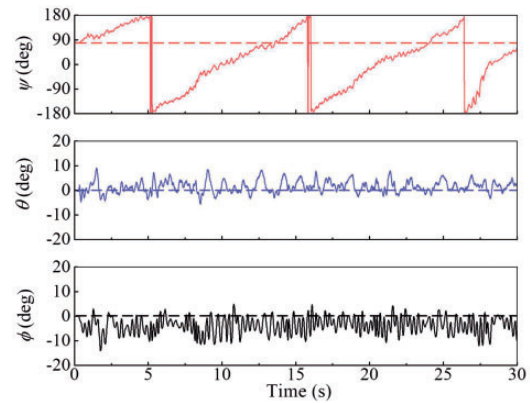


Figure 12. Attitude performance of KUBeetle-S during hovering flight without heading hold. Solid and dashed lines denote the body attitude angles and reference angles, respectively.

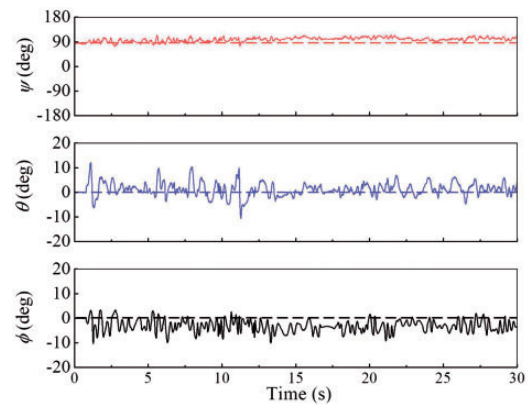


Figure 13. Attitude performance of KUBeetle-S during hovering flight with heading hold. Solid and dashed lines denote the body attitude angles and reference angles, respectively.

although slight oscillations around the reference angle occurred (Figure 13). In conclusion, the flight proves that the attitude control system is successfully implemented in an insect-like tailless flying robot.

Conclusion

This work introduces a 16.4 g insect-like tailless, two-winged, hover-capable KUBeetle-S robot that can change its stroke plane and wing twist simultaneously for pitch and roll controls and modulate wing root spars asymmetrically for yaw control. The proposed control mechanism, which requires less actuation torques from actuators, is simple and easy for fabrication allowing us to use small actuators to save weight. Measured data proved that the control mechanism could effectively generate reasonable amounts of force and torque for attitude controls. Finally, KUBeetle-S with the implementation of onboard attitude feedback control system successfully hovered and loitered for about 3 min, demonstrating the effectiveness of the proposed control mechanism.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: This research was partially supported by the 2018 KU Brain Pool Program of Konkuk University, Korea.

ORCID iD

Hoang Vu Phan  <https://orcid.org/0000-0002-4943-9765>

Hoon Cheol Park  <https://orcid.org/0000-0002-5070-6189>

References

1. Dudley R. *The biomechanics of insect flight: form, function, evolution*. Princeton, NJ: Princeton University Press, 2002.
2. Taylor GK. Mechanics and aerodynamics of insect flight control. *Biol Rev* 2001; 76: 449–471.
3. Zanker JM. On the mechanism of speed and altitude control in *Drosophila melanogaster*. *Physiol Entomol* 1988; 13: 351–361.
4. Ellington CP. The aerodynamics of hovering insect flight. I. The quasi-steady analysis. *Philos Trans R Soc B* 1984; 305: 1–15.
5. Vogel S. Flight in *Drosophila*. II. Variations in stroke parameters and wing contour. *J Exp Biol* 1967; 46: 383–392.
6. Srinivasan MV. A visually-evoked roll response in the housefly. Open-loop and closed-loop studies. *J Comp Physiol A* 1977; 119: 1–14.
7. Nachtigall W and Roth W. Correlations between stationary measurable parameters of wing movement and aerodynamic force production in the blowfly (*Calliphora vicina* R.-D.). *J Comp Physiol A* 1983; 150: 251–260.
8. Alexander DE. Unusual phase relationships between the forewings and hindwings in flying dragonflies. *J Exp Biol* 1984; 109: 379–383.
9. Burton AJ. Directional change in a flying beetle. *J Exp Biol* 1971; 54: 575–585.
10. Kammer AE. The motor output during turning flight in a hawkmoth *Manduca sexta*. *J Insect Physiol* 1971; 17: 1073–1086.
11. Park JH and Yoon KJ. Designing a biomimetic ornithopter capable of sustained and controlled flight. *J Bionic Eng* 2008; 5: 39–47.
12. de Croon GCHE, Groen MA, Wagter CD, et al. Design, aerodynamics and autonomy of the DelFly. *Bioinspir Biomim* 2012; 7: 025003.
13. Festo, SmartBird - Aerodynamic lightweight design with active torsion, https://www.festo.com/net/SupportPortal/Files/46270/Brosch_SmartBird_en_8s_RZ_110311_lo.pdf (accessed 2011).
14. Baek SS, Bermudez FLG and Fearing RS. Flight control for target seeking by 13 gram ornithopter. In: *IEEE conference on intelligent robots and systems (IROS)*, San Francisco, CA, USA, 25–30 September, 2011.
15. Nguyen QV, Chan WL and Debiase M. Hybrid design and performance tests of a hovering insect-inspired flapping-wing micro aerial vehicle. *J Bionic Eng* 2016; 13: 235–248.
16. Keennon MT, Klingebiel K, Won H, et al. Development of the Nano Hummingbird: a tailless flapping wing micro air vehicle. In: *50th AIAA aerospace sciences meeting*, Nashville, Tennessee, 9–12 January, 2012.
17. Phan HV, Kang TS and Park HC. Design and stable flight of a 21 g insect-like tailless flapping wing micro air vehicle with angular rates feedback control. *Bioinspir Biomim* 2017; 12: 036006.
18. Roshanbin A, Altartouri H, Karásek M, et al. Colibri: a hovering flapping twin-wing robot. *Int J Micro Air Veh* 2017; 9: 270–282.
19. Ma KY, Chirarattananon P, Fuller SB, et al. Controlled flight of a biologically inspired insect-scale robot. *Science* 2013; 340: 603–607.
20. Zhang J, Tu Z, Fei F, et al. Geometric flight control of a hovering robotic hummingbird. In: *IEEE conference on robotics and automation (ICRA)*, Singapore, 29 May – 3 June, 2017.
21. Karásek M, Muijres FT, de Wagter C, et al. A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns. *Science* 2018; 361: 1089–1094.
22. Coleman D, Benedict M and Chopra I. Design, development and flight testing of robotic hummingbird. In: *71st annual forum of the American Helicopter Society*, Virginia Beach, Virginia, USA, 5 – 7 May, 2015.
23. Nguyen QV and Chan WL. Development and flight performance of a biological-inspired tailless Flapping Wing Micro Air Vehicle with wing stroke plane modulation. *Bioinspir Biomim* 2019; 14: 016015.
24. Truong QT, Phan HV, Sane SP, et al. Pitching moment generation in an insect-mimicking flapping-wing system. *J Bionic Eng* 2014; 11: 36–51.

25. Phan HV and Park HC. Generation of control moments in an insect-like tailless flapping-wing micro air vehicle by changing the stroke-plane angle. *J Bionic Eng* 2016; 13: 449–457.
26. Phan HV and Park HC. Wing inertia as a cause of aerodynamically uneconomical flight with high angles of attack in hovering insects. *J Exp Biol* 2018; 221: jeb187369.
27. Phan HV, Au TKL and Park HC. Clap-and-fling mechanism in a hovering insect-like two-winged flapping-wing micro air vehicle. *R Soc Open Sci* 2016; 3: 160746.
28. Phan HV and Park HC. Design and evaluation of a deformable wing configuration for economical hovering flight of an insect-like tailless flying robot. *Bioinspir Biomim* 2018; 13: 036009.
29. Hedrick T. Software techniques for two- and three-dimensional kinematic measurements of biological and biomimetic systems. *Bioinspir Biomim* 2008; 3: 034001.
30. Le QT, Truong VT, Park SH, et al. Improvement of the aerodynamic performance by wing flexibility and elytra-hind wing interaction of a beetle during forward flight. *J R Soc Interface* 2013; 10: 20130312.
31. Phan HV, Truong QT, Au TKL, et al. Effect of wing kinematics modulation on aerodynamic force generation in hovering insect mimicking flapping-wing micro air vehicle. *J Bionic Eng* 2015; 12: 539–554.
32. Nguyen TA, Phan HV, Au TKL, et al. Experimental study on thrust and power of flapping-wing system based on rack-pinion mechanism. *Bioinspir Biomim* 2016; 11: 046001.
33. Aurecianus S, Phan HV, Nam SL, et al. Flapping wing micro air vehicle (FW-MAV) state estimation and control with heading and altitude hold. In: *10th international micro air vehicle competition and conference (IMAV)*, Melbourne, Australia, 17 – 23 November, 2018.